22MO0010
Empirical Methods
University of Zurich HS 2026
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Problem Set 3 and 4 – Replicating a Research Paper

The deliverables of these problem sets are due on the dates indicated below at **23:59**. They are to be handed in on OLAT. **Please follow the 'Instructions for Submission', which you find on OLAT. Problem Sets submitted with the wrong file name will not be graded!**

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1. Introduction

Replicating empirical research is an important part of good research practice. Only by replicating research can we learn whether important findings are correct and robust.

In this problem set, you will choose a paper that uses one of the methods covered in the course and replicate some of its main results. In addition, we will ask you to critically evaluate the paper based on the methods (e.g., OVB, simultaneous causality etc.) introduced in the course and then extend the analysis. Do the conclusions of the paper still hold when you use different (publicly available) data sources, alternative econometric methods, or different model specifications?

Importantly: We do not want you to reproduce the paper perfectly, but to understand the choices made by the authors and learn how to assess how much we can trust their results. These are the same skills you will need when reading empirical research and communicating its results to others. We also typically ask you to do something very similar in the exam, so learning how to evaluate papers and argue about econometric methods early on can become very valuable to you, at least in the short term.

2. The Task – Overview

- Problem Set 3: A 5 to 7-page replication report of a paper of your choice
- Problem Set 4: An interactive website about the paper

3. Getting Started: Find and Register a Paper (due 16.11.2026)

In the first step, you must find a paper of your choice. The paper can be from economics, finance, marketing, or any other related field of your interest. Importantly, it should use an econometric method covered in this course (i.e., OLS, Panel Data Methods, IV).

Please upload your group's chosen paper to the Empirical Methods platform (https://praxai.ch) by **16.11.2026**. The option to submit on the platform will become available after Thursdays TA session (29.10.2026). The Papers will be assigned on a first-come, first-served basis, and you will also be able to see which papers have already been selected by other groups. If another group has already chosen the same paper, you will need to select an alternative. So, start to coordinate early within your group. Additionally, please also indicate where you will get the original data from used in the paper (e.g., primary public sources or replication package shared by authors). Only papers with available data sources will be accepted. You must include a link to the data or the author's replication package when submitting your paper.

When selecting a paper, consider the following questions:

- Is the data publicly accessible, or do the authors make the data available?
- Are the necessary data cleaning (preprocessing) steps straightforward and replicable?
- Do you understand the identification strategy?
- Can you realistically replicate at least the main result of the paper?
- Are there opportunities for meaningful robustness checks or extensions?

Hint: A paper whose empirical strategy and data you understand is always preferable to a more complex one you can only partially grasp or reproduce. Most leading economics journals now require authors to publish data and code. Replication packages are archived at the American Economic Association (AER, AER: Insights, AEJs), the Econometric Society (Econometrica) and ReStud communities on Zenodo, and Harvard Dataverse (QJE, JPE, ReStat). You can search across journals at ejd.econ.mathematik.uni-ulm.de.

4. Problem Set 3: A Report with Replication Code (due 30.11.2026)

As a first deliverable, you will produce a replication report together with the code used for your analysis. The report should be structured into (at least) five distinct parts. We provide a template that you can use as a starting point and adjust as you see fit. It is not comprehensive nor limiting (e.g., you may add or remove sections), but it is intended to help structure your thinking when assessing the paper for the first time.

Start with the raw data provided by the authors and replicate the numbers of the paper as closely as possible. This involves cleaning the data appropriately, understanding which variables need to be included, and estimating the appropriate model. Try to identify where the differences between your and the authors' results come from. Don't aim for a perfect replication. Instead, focus on understanding the underlying theory and assumptions of the model and on correctly interpreting the main results you produce.

At a minimum, we ask you to cover the following:

(a) **Introduction:** What is the main problem identified by the authors? What is the main research question? Very briefly, what method do they use and what do they find? Also introduce your replication result and extension.

(b) **Methodology:** Which econometric method do the authors use to identify their main result? What assumptions does it rely on, and what are the general threats to this method?

(c) **Main Result:** Reproduce the main result of the paper. This includes identifying the main econometric specification and replicating at least the main result column of the coefficient table and one to two charts. Interpret the results: What can we learn from them, and how do they answer the research question? Why might your numbers not match the authors' exactly?

(d) **Critical Evaluation:** Do you see any threats to the internal validity of the authors' results? Use the methods discussed in class to explain and support your arguments. For example, if you make a claim about the omitted variable bias, use the formula discussed in class to support your intuition, and argue about the likely direction and size of the bias.

(e) **Extension and Robustness Check:** Come up with an extension to the paper. It can involve a different data source to check the robustness of the results, a different econometric method (including methods not discussed in the course), or an additional robustness check using different model specifications. What are the strengths of your extension? What could be improved? What are the limitations of the new approach? In addition, please think about an "ideal" empirical strategy: Which method would best answer the research question stated by the authors, and which data would you need to implement it? If you believe, for example, that an instrumental variable approach would improve the credibility of the results, argue why your proposed instrument is a good identification strategy and which data and model specification you would need to estimate it.

Please do not use AI to write your report (editing grammar using AI is, of course, encouraged). Being able to understand results and communicate them effectively is the most important skill of any researcher, and any obvious signs of over-reliance on AI will be considered when scoring the reports. We do encourage you to discuss your ideas and code with AI: this is how the profession is evolving, and collaborating with AI has become an increasingly valuable skill. You should, however, be able to critically evaluate and take responsibility for any output produced by AI. It is often more efficient to do simple tasks the "old way" (without AI) as opposed to iterating infinitely with AI until it "feels" like it was done the "old way."

We will not use any formal AI detection tools, but we will evaluate the reports as human experts, so your writing, analysis, and overall composition should communicate effectively to

us (humans!). We will not reward overly complicated (but convoluted) analysis or lengthy prose. Instead, we will reward signs of your own learning, honest effort, and clarity. If we fail to recognize them when grading, we hope that the universe (or the job market) will judge your honest efforts more accurately in the near future. :)

5. Problem Set 4: An Interactive Website (due 14.12.2026)

In addition to the report, you will build an interactive website that presents the main research question of your paper and communicates its main findings through 1–2 interactive visualizations. The website should be aimed at a general audience, so try to avoid technical jargon and focus on communicating the results clearly through the visualizations. For this deliverable, the use of AI is expected and encouraged for all the technical aspects. You may use HTML/CSS/JavaScript, Python/Streamlit, or any other approved web framework, and host the site for free on GitHub Pages, Streamlit Community Cloud, Vercel, or a similar provider (you should not spend any money for this project).

AI can be very helpful in explaining how these tools work, and we will discuss the setup in more detail, including an example, in one of the TA sessions.

We expect the following deliverables:

- A GitHub repository with the corresponding code. Either use a private repository and invite the TAs as collaborators, or use a public repository and share the link with the TAs.
- A publicly accessible website hosted on one of the free services mentioned above.
- Website content:
  - A paper summary box that explains the main research question and the main takeaways in non-technical language.
  - Intuitive headlines for every section.
  - At least one table (e.g., regression results) and one chart (visualizing descriptive statistics or regression results) with some dynamic functionality (e.g., filters, sliders, or toggle buttons that change the underlying sample, select different regression variables, display different categories, etc.).
  - Three exam-style questions below the charts that directly follow from your paper. Take a look at the mock exam for an example: our questions typically get progressively harder and challenge some of the main assumptions or model specifications used in the paper. Reading a paper and thinking about how it connects to the course is also how we start developing the main questions for the exam, so practicing how to formulate and answer such questions is excellent preparation.

Importantly, since the website link will be accessible to the public and you do not want anyone to see your names, you can host the website under a pseudonym (a team name). However, we would encourage you to make it personal: a nice project website is one of the best additions to your job market applications. Additionally, your website and especially the exam-like questions could serve as a useful learning resource for other students. We will ask each group individually, whether you would allow us to include your website link in an email to all other students in the course after the submission deadline.

Below are some reference examples. However, these are more extensive than what we expect for this course:

- `https://qgallea.github.io/gallea-rohner-interactive/`
- `https://www.role-model-effects.com/`